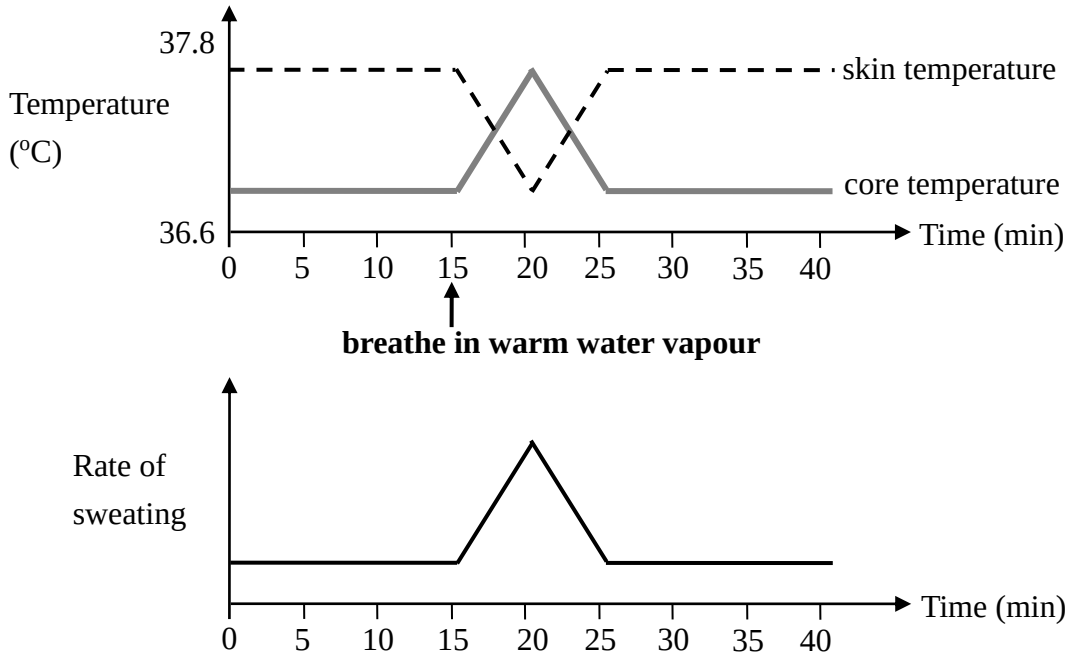
Structured questions

|!|BCELB0320000001|!| ★ **SI** **STSE**

In an experiment conducted at a constant environmental temperature of 25 °C, a man at rest was asked to breathe in 10 breaths of warm water vapour (about 45 °C). Measurements of the following parameters were taken from the man during the course of the experiment:

- core temperature
- skin temperature
- rate of sweating

The data are presented in the following graphs.



(a) Suggest ***two*** methods of how the core temperature of the man can be measured accurately. (2 marks)

_____

_____

(b) Describe and explain the changes in the man's core temperature and the skin temperature from the 15th to the 20th minutes. (5 marks)

_____

_____

_____

_____

© 2020 Aristo Educational Press Ltd.

(c) Explain why for patients suffering from hypothermia, breathing in warm water vapour is a more efficient first aid method to restore the body temperature than soaking in warm water. (3 marks)

##

(a) Measuring the temperature of the eardrum / the rectum / under the tongue / at the armpit (Any ***two***, 2)

(b) After breathing in warm water vapour, the blood in the lungs gained heat via conduction and radiation, so the temperature of the blood increased. (1)
The thermoreceptors in the hypothalamus detected an increase in the blood temperature. (1)
The heat loss centre in the hypothalamus sent nerve impulses to the skin / sweat glands to increase sweating. (1)
More heat was lost to the surroundings by evaporation of sweat. (1)
The skin temperature decreased as the rate of heat loss is higher than the rate of heat gain. (1)

(c) The lungs have a larger surface area than the skin. (1)
Thus the rate of heat gain by breathing in warm water vapour is higher than that by soaking in warm water. (1)
The body temperature of the patient will rise more rapidly to the normal range. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Analysing data	2
	Applying concepts	3
(c)	Applying concepts	3

© 2020 Aristo Educational Press Ltd.

|!|BCELA0320000002|!| ★★ **STSE**

In an experiment aiming to study the thermoregulatory responses, a man was asked to perform exercises in the following conditions:

Condition	Air temperature (°C)	Relative humidity (%)
A	35	40
B	35	90

(a) How does the man lose heat effectively under condition A? (1 mark)

(b) State and explain which condition is not suitable for the man to perform prolonged exercise. (6 marks)

(c) The Air Quality Health Index (AQHI) compiled by the Environmental Protection Department informs the public of the short-term health risk of air pollution in Hong Kong. When the index is very high, it is advised to reduce outdoor physical activities. Explain why. (2 marks)

##

(a) Through the evaporation of sweat (1)

(b) Condition B is not suitable. (1)

During exercise, heat is produced by muscle contractions. (1)

As the relative humidity is high, heat cannot be lost through evaporation of sweat. (1)

© 2020 Aristo Educational Press Ltd.

As the air temperature is high, heat cannot be lost effectively through conduction / convection / radiation. (1)

As a result, the rate of heat gain is higher than that of heat loss. (1)

The man may suffer from heatstroke / may become overheated if he performs prolonged exercise. (1)

(c) When performing physical activities, the ventilation rate / rate and depth of breathing increases. (1)

More pollutants will enter the lungs through inhalation. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
	Applying concepts	5
(c)	Applying concepts	2

|!|BCELB0320000003|!| ★

The rate of blood flow to various organs was measured when a teenager was at rest and during vigorous exercise.

(a) The rate of blood supply to the lungs increased during exercise. State the significance of the change. (2 marks)

(b) Besides the lungs, the rate of blood supply to the skin also changed during exercise.

(i) Suggest the change in the rate of blood supply to the skin during exercise. (1 mark)

(ii) Describe the regulatory mechanism that brings about the change in (b)(i). (4 marks)

© 2020 Aristo Educational Press Ltd.

(iii) Explain the significance of the change in (b)(i). (3 marks)

##

(a) Increasing the rate of blood supply to the lungs can transport blood to and away from the air sacs more quickly. (1)

This can create a steep concentration gradient of oxygen between the air sacs and the blood for the faster absorption of oxygen from the air in the air sacs into the blood. (1)

(b) (i) The rate of blood supply to the skin would increase during exercise. (1)

(ii) During exercise, heat is produced by muscle contractions. (1)

An increase in body temperature is detected by the thermoreceptors in the hypothalamus. (1)

The heat loss centre in the hypothalamus sends nerve impulses to the skin (1) to bring about vasodilation of the arterioles supplying blood to the capillaries near the skin surface. (1)

(iii) The increased blood flow brings more heat to the skin surface. (1)

Heat loss from the blood to the surroundings by conduction / convection / radiation increases. (1)

This helps prevent overheating / heatstroke. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	4
(b)(iii)	Applying concepts	3

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000004|!| ★ **STSE**

In a cold day, a man without enough clothing got lost in a forest.

(a) Give ***one*** behavioural response which the man can carry out to minimize heat loss. Explain how the behavioural response can minimize heat loss. (3 marks)

(b) Name ***one*** involuntary response observable in the man, which increases heat production. Describe the regulatory mechanism that brings about this response. (4 marks)

(c) Suggest ***two*** other physiological responses in the skin under cold conditions. (2 marks)

(d) Besides thermoregulation, the skin serves other functions which are important to our survival. Give ***one*** example. (1 mark)

##

(a) He can curl up this body. (1)
This posture decreases the surface area of the body exposed to cold air. (1)
Heat loss from the body surface to the surroundings by conduction / convection / radiation can be reduced. (1)

(b) Shivering (1)
A fall in body temperature is detected by the thermoreceptors in the hypothalamus. (1)
The heat gain centre in the hypothalamus sends more nerve impulses to the skeletal muscles (1)
causing the skeletal muscles to contract involuntarily in short bursts. (1)

© 2020 Aristo Educational Press Ltd.

(c) Vasoconstriction of arterioles supplying blood to the blood capillaries near the skin surface / contraction of hair erector muscles / decreased rate of sweating (Any ***two***, 2)

(d) Preventing the entry of pathogens / minimizing water loss through the body surface / excreting urea in sweat / sensation (Any ***one***, 1)

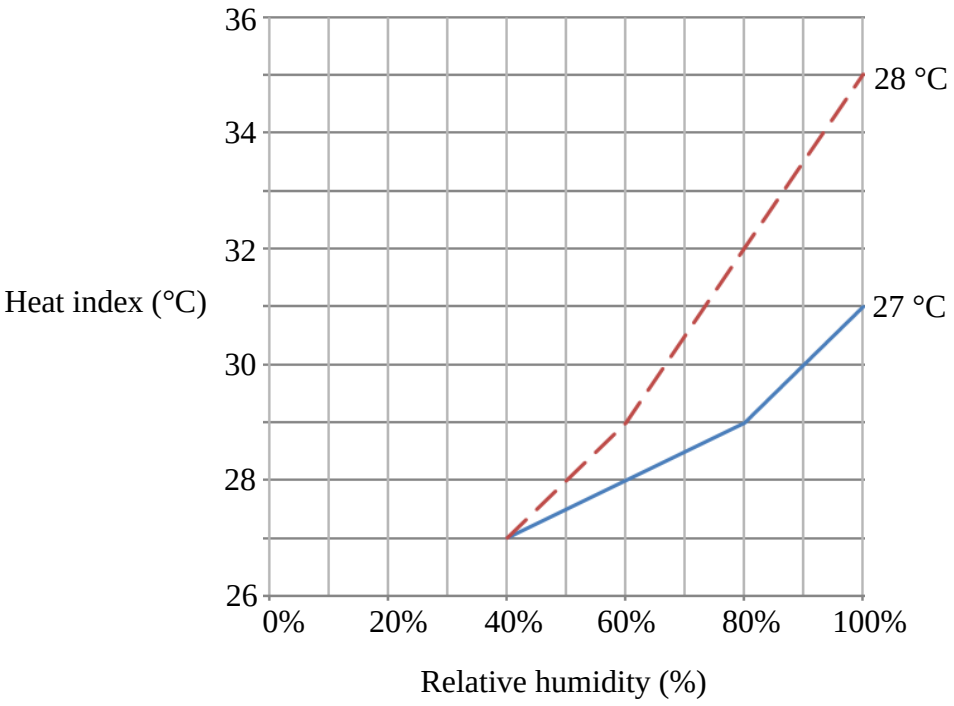
(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	4
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	1

|!|BCELA0320000005|!| ★★ **STSE**

The heat index reflects the temperature perceived by a person in a shaded area taking into account the air temperature and the relative humidity. The graph below shows the effect of relative humidity on the heat index when the air temperature is 27 ºC or 28 ºC respectively.



(a) With reference to the graph, what is the general pattern between increasing humidity and the heat index? (3 marks)

(b) Explain the effect of high relative humidity on the heat index. (3 marks)

(c) What is the main means of heat loss from the body when the relatively humidity is high? Describe the nervous coordination leading to this response. (3 marks)

(d) Besides relative humidity and air temperature, suggest another factor which could also affect how a person feels about the temperature of the environment. (1 mark)

##

(a) The heat index increases with increasing relative humidity starting from 40%. (1)
The extent of increase in the heat index is greater at higher relative humidity. (1)
At 28 °C, the heat index increases more rapidly with increasing relative humidity than at 27 °C. (1)

(b) At high relative humidity, the rate of sweat evaporation is very slow. (1)
Heat loss by sweat evaporation is ineffective. (1)
The rate of heat loss becomes lower than that of heat gain, (1)
so we feel hotter.

(c) Vasodilation of the arterioles supplying blood to the capillaries near the skin surface (1)
An increase in body temperature is detected by the thermoreceptors in the hypothalamus. (1)
The heat loss centre in the hypothalamus sends nerve impulses to the arterioles supplying blood to the capillaries near the skin surface (1)
in order to bring about vasodilation.

© 2020 Aristo Educational Press Ltd.

(d) Wind speed / light intensity / amount of clothing / physical activity (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	3
(c)	Applying concepts	1
	Understanding basic concepts	2
(d)	Understanding basic concepts	1

|!|BCELB0320000006|!| ★

Answer the questions below with respect to the following environmental conditions:

- Condition A: Hot and humid
- Condition B: Hot and dry
- Condition C: Cold and humid
- Condition D: Cold and dry

(a) (i) Under which of the above conditions should people stop doing intense physical exercise? (1 mark)

(ii) Explain your choice for (a)(i). (5 marks)

(b) When a person is in a hot environment, the person's face turns red. Describe the physiological response that leads to the redness of the face. (4 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) (i) Condition A (1)

(ii) During intense physical exercise, a large amount of heat is produced by the contraction of the skeletal muscles. (1)

When the environment is hot, it is difficult for the body to lose heat via conduction / convection / radiation. (1)

When the environment is humid, it is difficult for the body to lose heat via evaporation of sweat. (1)

As a result, the rate of heat gain would be higher than that of heat loss. (1)

The risk of overheating / heatstroke would be high. (1)

(b) When the environment is hot, the body temperature increases which is detected by the thermoreceptors in the hypothalamus. (1)

The heat loss centre in the hypothalamus sends nerve impulses to the skin (1)
to bring about vasodilation of the arterioles supplying blood to the blood capillaries near the skin surface. (1)

As a result, more blood flows near the skin surface (1)
and this results in the redness of the face.

(10 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Applying concepts	5
(b)	Applying concepts	4

|!|BCELB0320000007|!| ★

Answer the questions below with respect to the following environmental conditions:

- Condition A: Hot and humid
- Condition B: Hot and dry
- Condition C: Cold and humid
- Condition D: Cold and dry

(a) (i) Under which of the above condition would sweat be produced at the highest rate? (1 mark)

(ii) Explain your choice for (a)(i). (4 marks)

© 2020 Aristo Educational Press Ltd.

(b) When a person is in a cold environment, the person's face turns pale. Describe the physiological response that leads to the face turning pale. (4 marks)

##

(a) (i) Condition A (1)

(ii) When the environment is hot, it is difficult for the body to lose heat via conduction / convection / radiation. (1)

When the environment is humid, it is difficult for the body to lose heat via evaporation of sweat. (1)

As a result, the body temperature would increase which is detected by the thermoreceptors in the hypothalamus. (1)

The heat loss centre in the hypothalamus sends nerve impulses to the skin / sweat glands (1)

to increase sweat production.

(b) When the environment is cold, the body temperature decreases which is detected by the thermoreceptors in the hypothalamus. (1)

The heat gain centre in the hypothalamus sends nerve impulses to the skin (1)

to bring about vasoconstriction of the arterioles supplying blood to the blood capillaries near the skin surface. (1)

As a result, less blood flows near the skin surface. (1)

This makes the face look pale.

(9 marks)

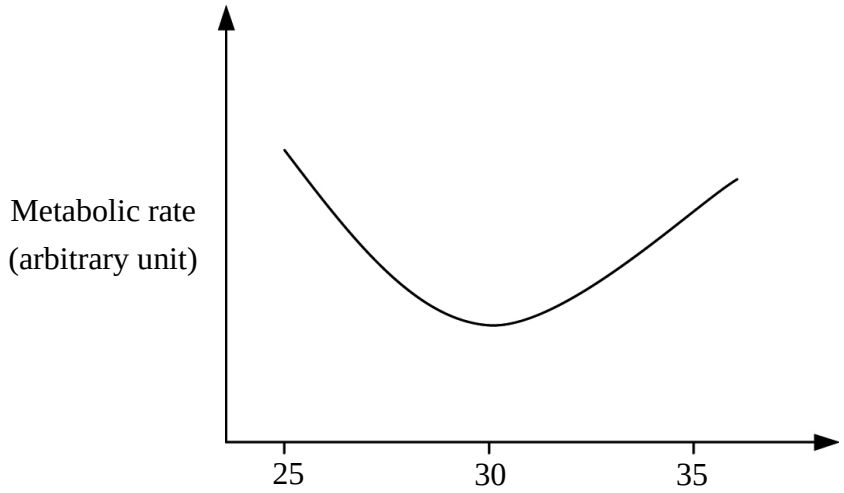
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Applying concepts	4
(b)	Applying concepts	4

© 2020 Aristo Educational Press Ltd.

|!|BCELA0320000008|!| ★★ **SI**

In an experiment, a naked man was asked to stand still in a temperature-controlled room. The temperature of the room was adjusted but its relative humidity was kept constant at 50%. The graph below shows how the metabolic rate of the man changes with air temperature.



(a) Describe and explain the change in metabolic rate when the air temperature decreases from 30 °C to 25 °C. (3 marks)

(b) Despite an increase in air temperature from 30 °C to 35 °C, the body temperature remains more or less constant. Describe the physiological responses that keep the body temperature stable. (4 marks)

(c) Explain why the life of the naked man in this experiment would be in danger if the air temperature rose to 40 °C and the relative humidity rose to 90%. (3 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) The metabolic rate increases. (1)
When the air temperature decreases, the rate of heat loss from the body by conduction / convection / radiation increases. (1)
An increased metabolic rate generates more heat to compensate for the heat lost. (1)

(b) Sweating is increased (1)
so that more heat can be lost from the skin by the evaporation of sweat. (1)
Vasodilation occurs / Arterioles in the skin dilates (1)
so that more blood flows through the capillaries near the skin surface and more heat can be lost by conduction / convection / radiation. (1)

(c) This is because heat loss by conduction, convection and radiation could not occur from 37 ºC to 40 ºC / against the temperature gradient. (1)
In addition, the heat lost by the evaporation of sweat would be reduced significantly when the relative humidity rose to 90%. (1)
As a result, the body would be overheated / enzymes in the body would be denatured. (1)

(10 marks)

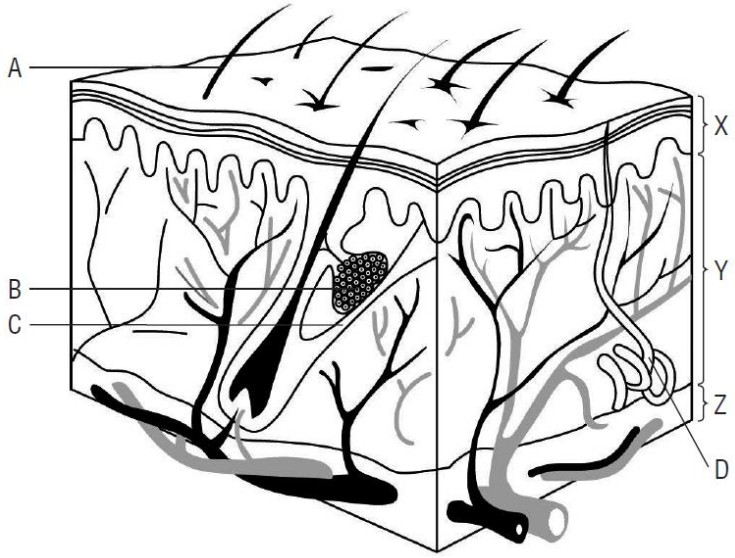
##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Applying concepts	4
(c)	Applying concepts	3

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000009|!| ★

The diagram below shows the structure of the mammalian skin.



(a) Name layers X and Y. (2 marks)

(b) Give ***two*** functions other than temperature regulation carried out by layer X. (2 marks)

(c) Describe how structures A and C respond when the air temperature decreases? (3 marks)

(d) Which structure is most effective in regulating body temperature when the air temperature rises above the body temperature? Explain your answer. (3 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) Layer X is the epidermis. (1)

Layer Y is the dermis. (1)

(b) Any ***two*** of the following: (2)

- It provides physical protection / prevents invasion of pathogens / prevents water loss.
- It produces vitamin D.

(c) Structure C contracts (1)

and pulls the base of structure A, therefore structure A becomes erect. (1)

A thicker layer of air is trapped among structure A. (1)

(d) Structure D (1)

When the air temperature is higher than the body temperature, the body can only lose heat to the surroundings by the evaporation of sweat. (1)

but not by conduction, convection and radiation when it is against the temperature gradient. (1)

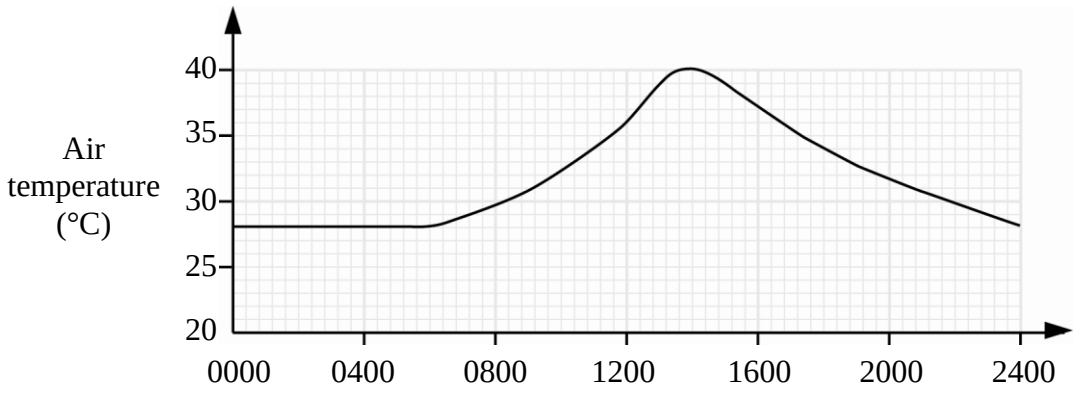
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	3
(d)	Analysing data	1
	Applying concepts	2

|!|BCELB0320000010|!| ★ **STSE**

The graph below shows the changes in the air temperature on a hot summer day.



A constructor worked outdoor on that day from 0800 to 1600.

© 2020 Aristo Educational Press Ltd.

(a) Sketch on the graph above to show the change in his core temperature during that day. (1 mark)

(b) Describe the physiological changes in the blood vessels in his skin from 0800 to 1200. Explain how the changes help the regulation of body temperature. (5 marks)

(c) The relative humidity on that day was 85%. The constructor had a high risk of having heatstroke.

(i) Suggest why the constructor had a high risk of having heatstroke. (3 marks)

(ii) Suggest a way to reduce the risk. (1 mark)

##

(a) the curve should stay at approximately 37 ^{o}C with slight fluctuations (1)

(b) The air temperature increases during 0800 to1200. Heat gain of the body increases during this period. (1)
The hypothalamus detects an increase in body temperature and sends nerve impulses to the skin. (1)
The arterioles near the skin surface dilate. (1)
More blood flows through the capillaries near the skin surface. (1)
More heat is lost from the skin to the surroundings by conduction, convection and radiation. (1)

(c) (i) When the air temperature is high, heat loss from the body by conduction, convection and radiation becomes less effective. (1)
When the humidity is high, the rate of evaporation of sweat decreases. (1)
There is net heat gain by the body, and this may lead to over-heating of the body. (1)

© 2020 Aristo Educational Press Ltd.

(ii) Drink plenty of water / rest under shaded area / other reasonable answer (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	1
	Applying concepts	4
(c)(i)	Applying concepts	3
(c)(ii)	Understanding basic concepts	1

|!|BCELC0320000011|!|

Ben participated in a long-distance race. His body temperature increased when he was running.

(a) Explain the rise in his body temperature. (3 marks)

(b) Describe and explain how the following structures help regulate his body temperature during the race.

(i) sweat glands (3 marks)

(ii) arterioles in the skin (3 marks)

##

(a) During running, the energy demand of skeletal muscles increases because of increased muscle contractions. (1)

© 2020 Aristo Educational Press Ltd.

The rate of respiration increases to supply more ATP. Some of the energy is released as heat. (1)

When heat gain is greater than heat loss of the body, the body temperature increases. (1)

(b) (i) The sweat glands secrete more sweat. (1)

Evaporation of sweat from the skin surface (1)

absorbs heat from the skin. (1)

(ii) The arterioles in the skin dilate. (1)

More blood flows through the capillaries near the skin surface. (1)

More heat is lost from the blood to the surroundings by conduction, convection and radiation. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)(i)	Understanding basic concepts	3
(b)(ii)	Understanding basic concepts	3

|!|BCELB0320000012|!| ★

Kelvin and Kenny are identical twins. Both of them are healthy and have similar metabolic rates. They were asked to stay in two identical rooms respectively for six hours. The rooms had the same environmental conditions, except that the room Kelvin stayed in was drier. Exercise, eating and drinking were not allowed before and during the experiment. Each of them just sat on a chair. They did not urinate or defaecate during the experiment.

(a) Who had a higher rate of water loss throughout the experiment? Explain. (2 marks)

(b) Suggest ***two*** ways by which the twins lost water during the experiment. (2 marks)

(c) If they had performed vigorous exercise during the experiment, explain why the water loss from the body would have increased. (6 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) Kelvin (1)

The rate of water loss by evaporation was higher in a drier room. (1)

(b) sweating (1)

exhalation (1)

(c) During vigorous exercise, the increased muscle contractions would produce more heat, leading to an increase in the body temperature. (1)

An increase in body temperature stimulates the heat loss centre in the hypothalamus. (1)

The heat loss centre sends more nerve impulses to the sweat glands to increase sweating. (1)

More water is lost in sweat. (1)

During vigorous exercise, the rate and depth of breathing would increase. (1)

More water vapour would be lost in exhaled air. (1)

(10 marks)

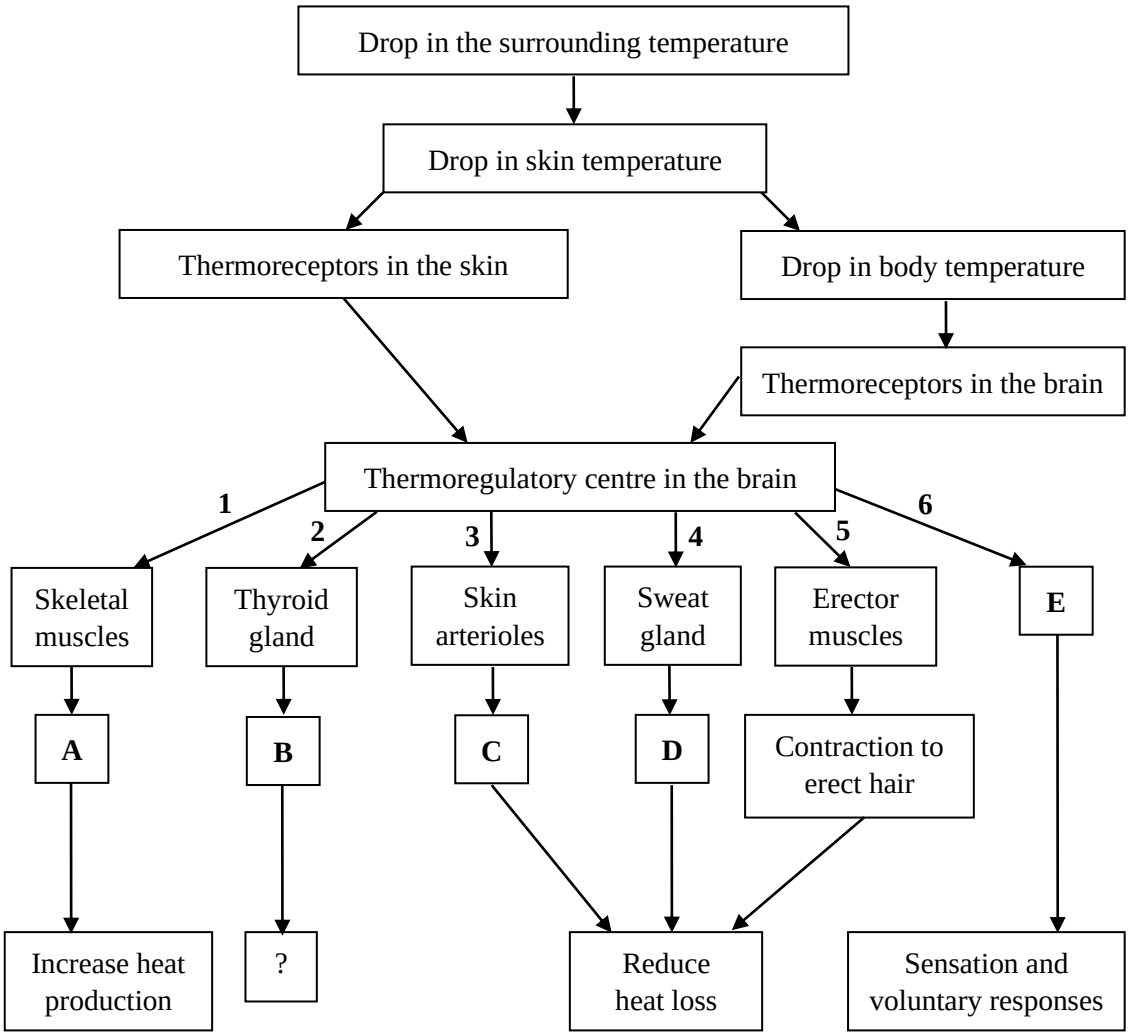
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	6

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000013|!| ★

The following flowchart shows the various nervous pathways in a person when the surrounding temperature decreases.



(a) Response A is involuntary. State this response carried out by skeletal muscles and explain how it increases heat production of the body. (3 marks)

(b) State response B and its effect in regulating the body temperature. (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) (i) State responses C and D. (2 marks)

(ii) Explain how responses C and D lead to reduction in heat loss. (3 marks)

(d) Pathway 6 results in sensation and voluntary responses initiated by structure E in the body.

(i) Name structure E. (1 mark)

(ii) Suggest ***one*** voluntary response under cold condition. (1 mark)

##

(a) Shivering (1)
Skeletal muscles contract in short bursts. (1)
Increased cellular respiration in the skeletal muscles generates more heat. (1)

(b) Secretion of more thyroxine (1)
It can increase the metabolic rate. (1)
The rate of heat production from metabolism is increased, which can compensate for heat loss. (1)

(c) (i) Response C: Vasoconstriction of the skin arterioles (1)
Response D: Reduced sweating (1)

(ii) Due to response C, less blood flows through the capillaries near the skin surface. (1)
This reduces heat loss from the blood to the surroundings by conduction, convection and radiation. (1)
Due to response D, heat loss through the evaporation of sweat is reduced. (1)

(d) (i) Cerebral cortex / cerebrum (1)

(ii) Any ***one*** of the following: (1)
- curling up the body
- wearing more clothes
- doing exercise

(Or other reasonable answer)

© 2020 Aristo Educational Press Ltd.

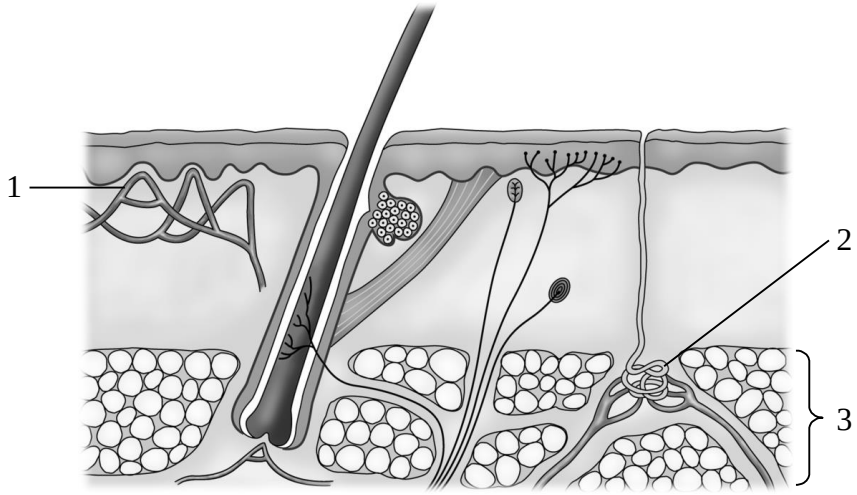
(13 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	3
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	3
(d)(i)	Understanding basic concepts	1
(d)(ii)	Applying concepts	1

|!|BCELC0320000014|!|

The following diagram shows a longitudinal section of the human skin.



(a) Name structure 1 and explain how it can facilitate heat loss under hot conditions. (5 marks)

(b) Name the fluid secreted by structure 2 and explain how this fluid can facilitate heat loss under hot conditions. (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) What is the difference in structure 3 between a person who lives in tropical region and a person who lives in polar region? (1 mark)

_____

_____

##

(a) Structure 1 is a blood capillary. (1)
Under hot condition, the arterioles in the skin dilate (1)
so that more blood flows through structure 1. (1)
Structure 1 is close to the body surface. (1)
More heat can be lost from the blood to the surroundings by conduction, convection and radiation. (1)

(b) The fluid secreted by structure 2 is sweat. (1)
Evaporation of sweat cools the body (1)
by absorbing heat from the skin. (1)

(c) Structure 3 is thinner in a person who lives in tropical region than a person who lives in polar region. (1)

(9 marks)

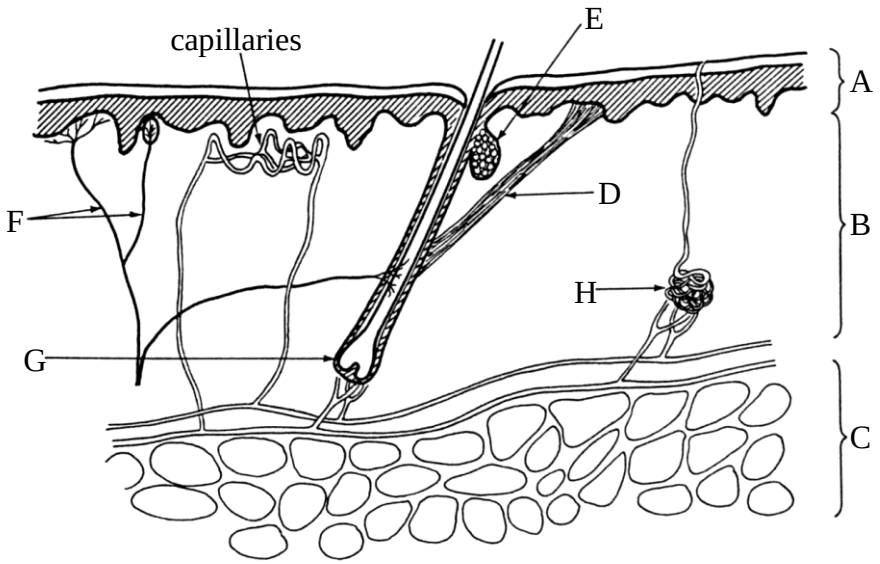
##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	4
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCELC0320000015|!|

The following diagram shows a longitudinal section through the human skin.



(a) Name the labelled structure in which a skin pigment is formed. Name the skin pigment and state ***one*** of its functions. (3 marks)

(b) State and explain ***one*** way in which the skin can help in regulating the body temperature

(i) when we feel cold. (2 marks)

(ii) when we feel hot. (2 marks)

(c) (i) Name the secretion of structure E. (1 mark)

(ii) State ***one*** function of this secretion. (1 mark)

##

(a) Structure A (1)

The skin pigment is melanin. (1)

Melanin determines the skin colour / absorbs the harmful ultraviolet (UV) light from the sun. (1)

(b) (i) Any ***one set*** of the following: (2)

- The arterioles in the skin constrict, so less blood flows through the capillaries near the skin surface. (1)
  Less heat is lost from the blood to the surroundings. (1)
- The sweat glands secrete less sweat. (1)
  Less heat is lost by evaporation of sweat. (1)

(Or other reasonable answers)

(ii) Any ***one set*** of the following: (2)

- The sweat glands secrete more sweat. (1)
  The evaporation of sweat at the skin surface cools the body by absorbing heat from the skin. (1)
- The arterioles in the skin dilate, so more blood flows through the capillaries near the skin surface. (1)
  More heat is lost from the blood to the surroundings by conduction, convection and radiation. (1)

(Or other reasonable answers)

(c) (i) sebum (1)

(ii) It keeps the skin soft and makes it waterproof.
OR
It can inhibit the growth of bacteria and fungi on the skin. (1)

(9 marks)

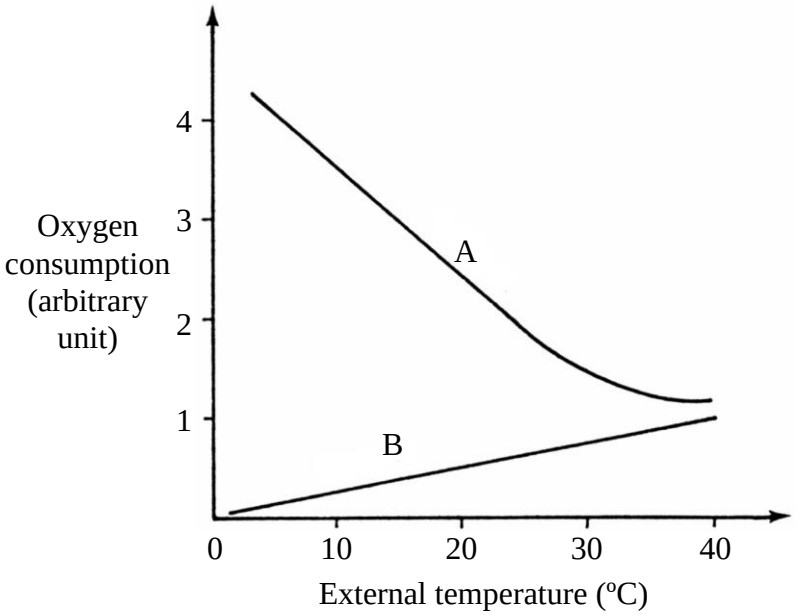
##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCELA0320000016|!| ★★

The graph below shows the changes in the oxygen consumption of a mouse and a frog in response to an increase in the external temperature.



(a) (i) Which curve represents the oxygen consumption of the mouse? (1 mark)

(ii) Explain your answer in (i). (2 marks)

(iii) Explain why the mouse is called a homoiotherm. (1 mark)

(b) The mouse is put into an insulated container and it is fastened so that it cannot move. If the temperature of the container is gradually lowered from 25 ºC to 15 ºC, how will the oxygen consumption of the mouse change? Explain your answer. (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) Explain the shape of curve B. (3 marks)

##

(a) (i) Curve A (1)

(ii) At lower temperatures, the mouse has to produce more heat to maintain a constant body temperature. (1)

Its respiration rate increases to release more heat and more oxygen is consumed. (1)

(iii) It can keep its body temperature constant despite fluctuations in the environmental temperature. (1)

(b) The oxygen consumption by the mouse will increase due to shivering. (1)

When the environment is cold and the body temperature of the mouse falls, it shivers involuntarily. (1)

Its skeletal muscles contract in short bursts and this releases more heat to compensate for the heat loss. (1)

(c) As the external temperature increases, the body temperature of the frog increases. (1)

Enzyme activity increases accordingly. (1)

The rate of metabolism and thus the oxygen consumption increase. (1)

(10 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Applying concepts	2
(a)(iii)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	3

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000017|!| ★ **STSE**

Tony learns that when a person is taking a cold shower, the blood flow through the capillaries near the skin surface reduces.

(a) Describe how the above response is resulted. (5 marks)

(b) Tony suggests taking a cold shower after vigorous exercise to help the body cool down. Comment on whether this suggestion works. (4 marks)

##

(a) Cold water removes heat from the skin, so the skin temperature decreases. (1)
The thermoreceptors in the skin detect a fall in the skin temperature. (1)
The heat gain centre in the hypothalamus sends nerve impulses to the skin arterioles. (1)
The skin arterioles constrict. (1)
More blood flows through the shunt vessels. (1)
As a result, the blood flow to the capillaries near the skin surface is reduced.

(b) No. (1)
Taking a cold shower would lead to a reduced amount of blood flowing through the capillaries near the skin surface. (1)
The reduced blood flow near the skin surface decreases the heat loss by conduction, convection and radiation. (1)
Therefore, body heat cannot be lost effectively and this may lead to overheating. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	5
(b)	Applying concepts	4

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000018|!| ★ **STSE**

Kevin goes cycling outdoors every morning.

(a) On a cool and dry day, Kevin goes cycling as usual.

(i) Suggest ***two*** ways by which heat is lost from his body. (2 marks)

(ii) Explain how a constant body temperature at around 37 ºC can allow cells to function properly. (3 marks)

(b) On a sunny and humid day, the air temperature reaches 42 ºC. Kevin goes cycling as usual.

(i) Suggest ***two*** ways by which he gains heat. (2 marks)

(ii) After cycling for a while, Kevin feels unwell and is later sent to the hospital. The doctor finds that his body temperature reaches 40.2 ºC. However, he does not have any infection and has drunk enough water before and during cycling. Explain the rise in his body temperature. (6 marks)

##

(a) (i) Any ***two*** of the following: (2)

- evaporation of sweat
- exhalation
- drinking cool water

© 2020 Aristo Educational Press Ltd.

- heat transfer from the skin surface to the surroundings by conduction, convection and radiation

(Or other reasonable answers)

(ii) Most metabolic reactions in our body are catalysed by enzymes. (1)

Many enzymes in the human body work best at an optimum temperature of about 40 ºC (near the body temperature of 37 ºC). (1)

A body temperature of 37 ºC ensures that the rate of metabolic reactions can support the proper functioning of the body. (1)

(b) (i) Any ***two*** of the following: (2)

- muscle contraction
- cell metabolism
- radiation from the sun
- heat transfer from the hotter air to the body surface by conduction and convection

(Or other reasonable answers)

(ii) During vigorous exercise, the respiration rate of the skeletal muscles increases in order to supply more energy for skeletal muscle contraction. (1)

This leads to the release of more body heat. (1)

The surrounding temperature is higher than the body temperature and so heat is gained from the surroundings. (1)

No heat is lost by conduction, convection and radiation. (1)

Also, the high humidity slows the rate of sweat evaporation and hence reduces heat loss from the body. (1)

As a result, the rate of heat gain is higher than the rate of heat loss. (1)

(13 marks)

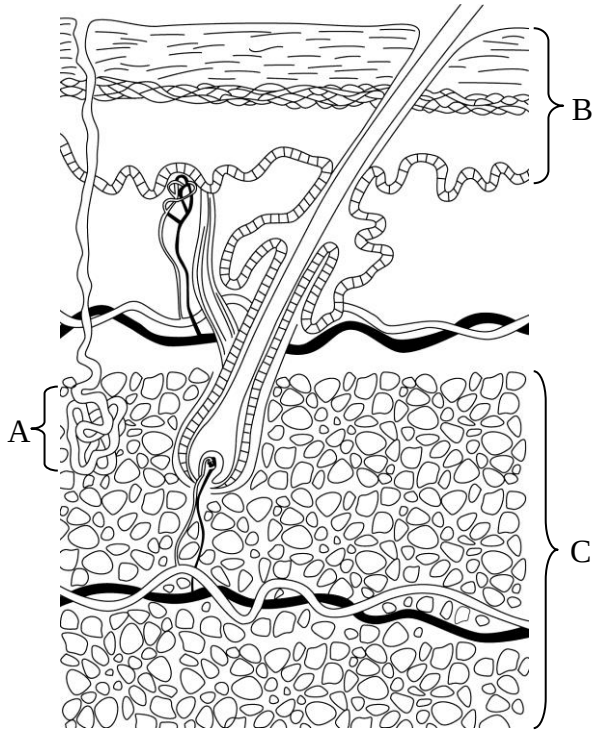
##

	Skills assessed	Marks
(a)(i)	Applying concepts	2
(a)(ii)	Applying concepts	3
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	6

© 2020 Aristo Educational Press Ltd.

|!|BCELC0320000019|!|

(a) The diagram below shows a longitudinal section of the human skin.

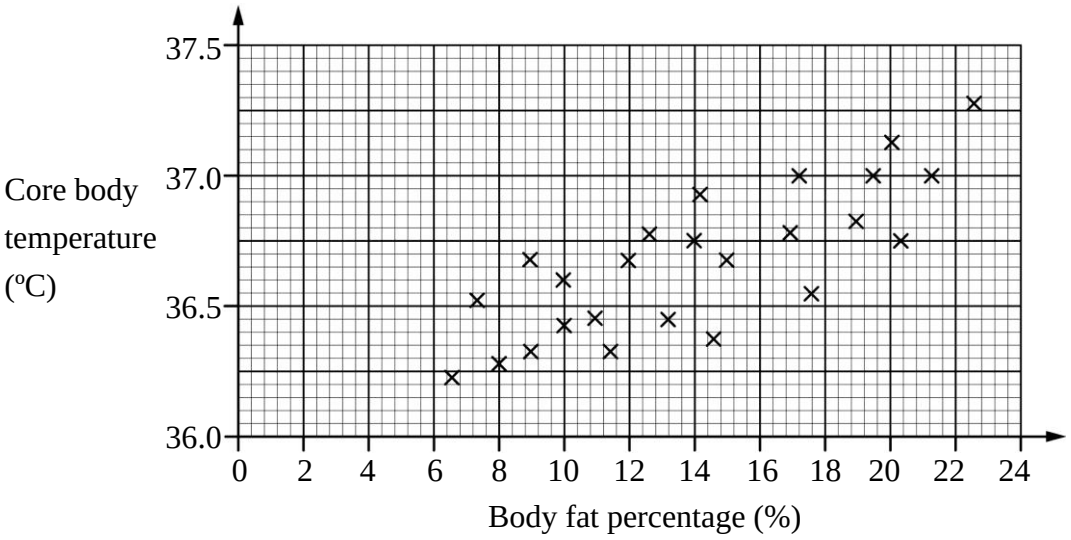


(i) Name the structures labelled A, B and C. (3 marks)

(ii) Explain the role of structure A in the regulation of body temperature under hot condition. (3 marks)

(b) In an investigation, 25 naked men were asked to sit in a room that was maintained at an air temperature of 15 °C. After two hours, their core body temperatures were measured. Their body fat percentages were also measured. The following graph shows the results of this investigation.

© 2020 Aristo Educational Press Ltd.



(i) According to the graph, state the relationship between body fat percentage and core body temperature. (1 mark)

(ii) Suggest explanations for the results shown. (2 marks)

##

(a) (i) A: sweat gland (1)
B: epidermis (1)
C: subcutaneous fat (1)

(ii) More sweat is secreted by structure A as the body temperature increases. (1)
Evaporation of sweat provides a cooling effect / brings a lot of body heat away from the skin surface. (1)
The body temperature falls and returns to normal. (1)

(b) (i) The core body temperature increases with an increase in body fat percentage. (1)

(ii) As fat is a good insulator of heat / provides insulation, (1)
more body fat results in less heat loss by conduction or radiation / more heat conservation. (1)

(9 marks)

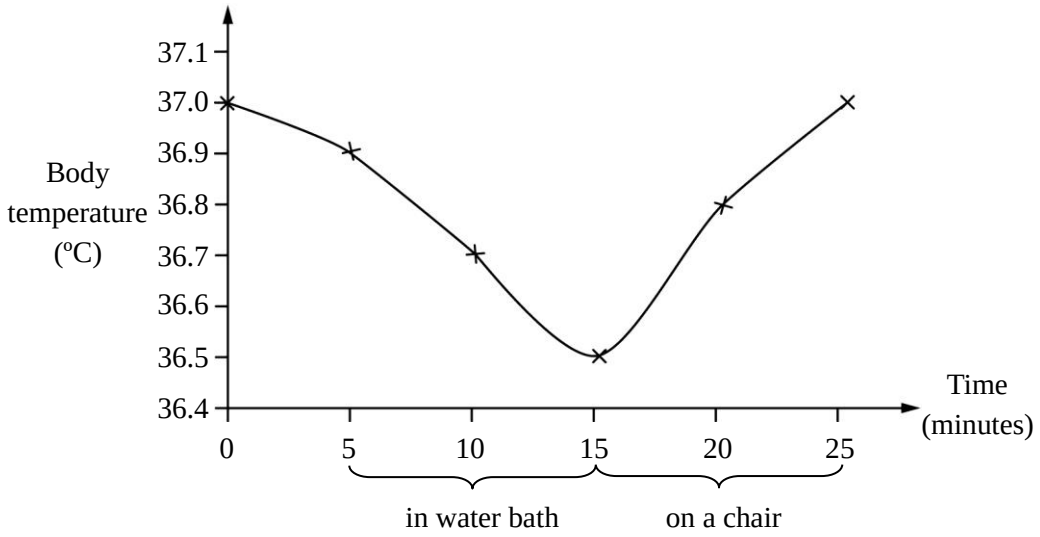
© 2020 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)(i)	Analysing data	3
(a)(ii)	Understanding basic concepts	3
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	2

|!|BCELB0320000020|!| ★ **SI**

In an investigation, Jimmy was first sat on a chair for five minutes. Then, he got into a water bath at a temperature of 20 ºC and stayed in the bath for ten minutes. After this, he left the bath and sat on a chair. During the investigation, his body temperature was recorded at regular time intervals. The following graph shows the results.



(a) Describe the changes in his body temperature during the investigation. (2 marks)

_____

_____

(b) Suggest explanations for the changes in his body temperature during the period of

(i) 5 to 10 minutes. (2 marks)

_____

_____

(ii) 15 to 25 minutes. (6 marks)

_____

© 2020 Aristo Educational Press Ltd.

##

(a) From 0 to 15 minutes, his body temperature decreased. (1)

From 15 to 25 minutes, his body temperature increased. (1)

(b) (i) The temperature of the water bath (20 ºC) was lower than his body temperature (37 ºC). (1)

Heat was lost from his body to the surrounding water by conduction. (1)

(ii) Thermoreceptors in the skin / hypothalamus detect a fall in the skin / blood temperature. (1)

The heat gain centre in the hypothalamus then initiates responses that increase heat gain and reduce heat loss. (1)

Shivering / Contraction of skeletal muscles generates more heat. (1)

The arterioles near the skin surface constrict, reducing blood flow to the capillaries near the skin surface. / Sweat glands secrete less sweat. (1)

Less heat is lost from the skin to the surroundings by conduction / convection / radiation / evaporation of sweat. (1)

The rate of heat gain is higher than the rate of heat loss. (1)

(10 marks)

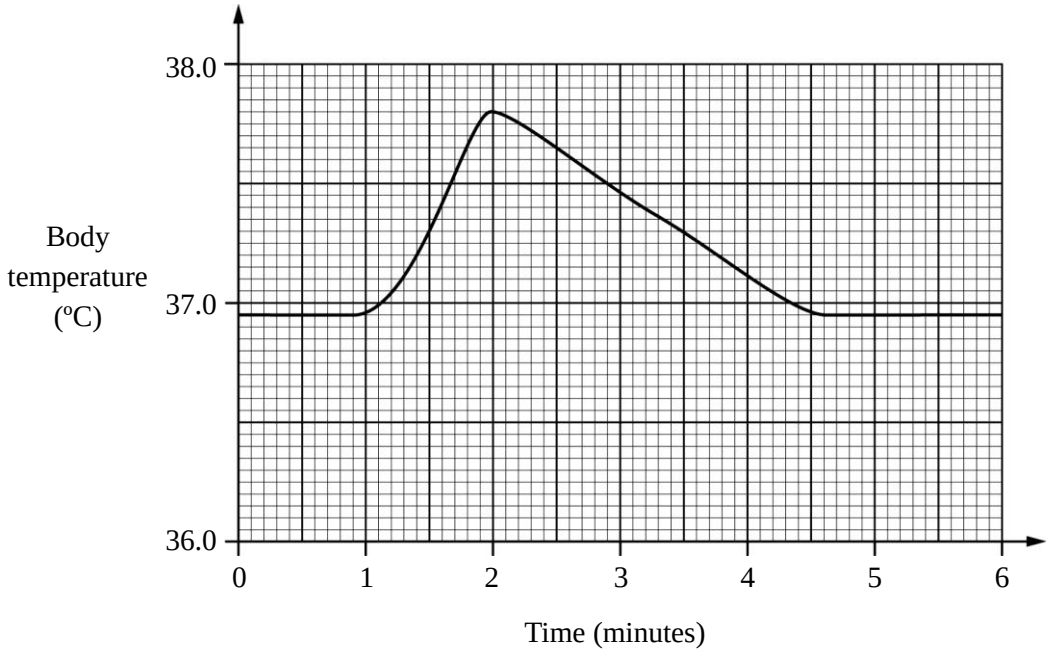
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	6

© 2020 Aristo Educational Press Ltd.

|!|BCELC0320000021|!|

The graph below shows the changes in Judy's body temperature after she drank a bowl of hot soup.



(a) Which part of the body can detect changes in the body temperature? (1 mark)

(b) How long did it take for the body temperature to return to the normal level from the maximum level? (1 mark)

(c) Describe the regulatory mechanism that brings the body temperature back to the normal level. (7 marks)

##

(a) hypothalamus (1)

(b) 2.5 minutes (1)

© 2020 Aristo Educational Press Ltd.

(c) In the alimentary canal, the heat of the soup is absorbed by the blood, so the blood temperature increases. (1)

The hypothalamus detects the increase in the blood temperature and sends nerve impulses to the skin. (1)

The skin arterioles dilate. (1)

This increases the rate of blood flow through the capillaries near the skin surface. (1)

More heat is lost by conduction, convection and radiation. (1)

More sweat is secreted by the sweat glands. (1)

Evaporation of sweat also cools the body by absorbing heat from the skin. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	1
(c)	Applying concepts	7

|!|BCELB0320000022|!| ★ **STSE**

A fund-raising marathon race has to be postponed due to bad weather. The teacher-in-charge has collected weather information of the following three days for consideration.

Date	Maximum temperature (°C)	UV index
14 June	29	> 11 (Extreme)
16 June	32	3–5 (Moderate)
18 June	28	6–7 (High)

(a) If you were the teacher-in-charge, what other weather information you think would be useful for reference as well? Explain your answer. (4 marks)

(b) Based on the information in the table only, which date has the lowest risk of heat stroke for the runners. Explain your choice. (4 marks)

© 2020 Aristo Educational Press Ltd.

(c) As a contingency plan, suggest ***two*** treatments for runners suffering from heat stroke. (2 marks)

##

(a) Relative humidity (1)
Relative humidity affects the rate of evaporation of sweat (1)
which in turn affects the rate of heat loss from the skin / body. (1)
It is an important factor for the risk of heat stroke. (1)

(b) 18 June (1)
The temperature on this day is lower than the other two days. (1)
The temperature difference (gradient) between the skin and air is larger. Therefore, more heat can be lost through conduction / convection / radiation. (1)
UV index has negligible / little effect on the rate of heat loss from the skin. (1)

(c) Move the runner to a shaded and cool place. (1)
Give cool mineral water to the runner. (1)
(Or other reasonable answers)

(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Analysing data	1
	Applying concepts	3
(c)	Understanding basic concepts	2

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000023|!| ★

Eskimos living in the cold Arctic region need to wear thick clothing which retains body heat as it is life threatening when the body temperature drops below a certain point.

(a) (i) Name the physiological condition when the body temperature drops to 35 ºC or below. (1 mark)

_____

(ii) State ***two*** symptoms shown by a person who has this physiological condition. (2 marks)

_____

_____

(b) Describe how homeostasis helps people living in cold climate maintain a constant body temperature despite the surrounding temperature is very low. (6 marks)

_____

_____

_____

_____

_____

_____

##

(a) (i) hypothermia (1)

(ii) Any ***two*** of the following: (2)

- pale skin
- dry skin
- slow pulse rate
- shivering
- reduced consciousness / faint

(Or other reasonable answers)

(b) Under cold condition, the body temperature decreases when the rate of heat loss is higher than the rate of heat gain. (1)

Thermoreceptors in the skin are stimulated and send nerve impulses to the hypothalamus. (1)

The heat gain centre is stimulated to initiate responses that reduce heat loss and increase heat gain. (1)

Any ***three*** of the following responses: (3)

© 2020 Aristo Educational Press Ltd.

- Erector muscles contract, therefore the hairs erect. This can trap a thicker layer of still air which provides insulation.
- Vasoconstriction of skin arterioles causes less heat loss from the blood near the skin surface.
- Shivering generates more heat.
- Thyroid gland secretes more thyroxine to increase the metabolic rate and hence the rate of heat production.

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)	Applying concepts	6

|!|BCELB0320000024|!| ★ **Cross topic**

The table below shows the amount of water lost in sweat and in urine from a person on two different days. He was at rest on a day and did vigorous exercise on another day.

	Amount of water lost (cm^3)	
	On rest day	**On exercise day**
Sweat	600	3500
Urine	1500	1000

(a) Explain the significance of increase in sweat production during vigorous exercise. (2 marks)

(b) Other than the increase in sweat production, describe ***one*** response in the skin that helps to maintain a constant body temperature during vigorous exercise. (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) Describe and explain the difference in the amount of water lost in urine on the two days. (6 marks)

##

(a) The body generates more heat during exercise. (1)
Increased sweating helps increase heat loss from the body by evaporation of sweat. (1)

(b) The arterioles in the skin dilate. (1)
More blood flow to the capillaries near the skin surface. (1)
More heat is lost to the surroundings through conduction, convection and radiation. (1)

(c) The amount of water lost in urine is smaller on the exercise day than on the rest day. (1)
More water is lost by sweating during vigorous exercise. (1)
The water potential of the blood falls. (1)
More ADH is released by the pituitary gland. (1)
The increased ADH level causes the wall of the collecting duct to become more permeable to water. (1)
Thus a greater proportion of water is reabsorbed by the collecting ducts of the kidneys. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	3
(c)	Analysing data	1
	Applying concepts	5

© 2020 Aristo Educational Press Ltd.

|!|BCELB0320000025|!| ★ **Cross topic** **STSE**

The table below shows the composition of sweat secreted by a person.

	Nitrogenous compounds	**Sodium chloride**	**Glucose**
Amount of constituents (μg / 100 cm^3)	32.0	4.5	2.3

(a) Name ***one*** of the nitrogenous compounds. (1 mark)

(b) State and explain the effects of sweating on urination during a hot day. (6 marks)

(c) Some people use a kind of spray (antiperspirant) that blocks the sweat ducts to prevent sweating. Explain why antiperspirant should not be sprayed all over the body when we use it. (3 marks)

(d) People may have body odour when they sweat a lot and do not wash regularly. Explain why the regular use of an antibacterial soap is better than an antiperspirant spray for controlling body odour. (2 marks)

##

(a) urea / other reasonable answer (1)

© 2020 Aristo Educational Press Ltd.

(b) During a hot day, more sweat is secreted and the water potential of the blood falls. (1)
The fall in water potential is detected by osmoreceptors in the hypothalamus. (1)
The hypothalamus stimulates the pituitary gland to release more ADH. (1)
The increased ADH level causes the wall of the collecting duct to become more permeable to water. (1)
A larger proportion of water is reabsorbed into the blood. (1)
As a result, a smaller volume urine is produced. (1)

(c) If antiperspirant is sprayed all over the body, sweating is inhibited. (1)
Heat loss by the evaporation of sweat is reduced. (1)
The body would overheat and this affects metabolism (as enzyme may be denatured at high temperature). (1)

(d) Body odour is caused by the breakdown of nitrogenous compounds / waste products by bacteria. (1)
Antibacterial soap can kill the bacteria and reduce the body odour. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	6
(c)	Applying concepts	3
(d)	Applying concepts	2

© 2020 Aristo Educational Press Ltd.